PhonePe Payment Insights Dashboard

Power BI Project Case Study | Interactive, Slicer-Driven Dynamic Insights

Project Overview

This dashboard analyzes ~300K simulated PhonePe transactions to surface payment behavior across services, age segments, and time. Beyond static charts, the report was built with **DAX-driven, filter-aware insight text** that updates live as the user interacts with the Month and Payment Status (Failed / Pending / Successful) slicers — turning a standard KPI dashboard into a self-narrating analytics tool.

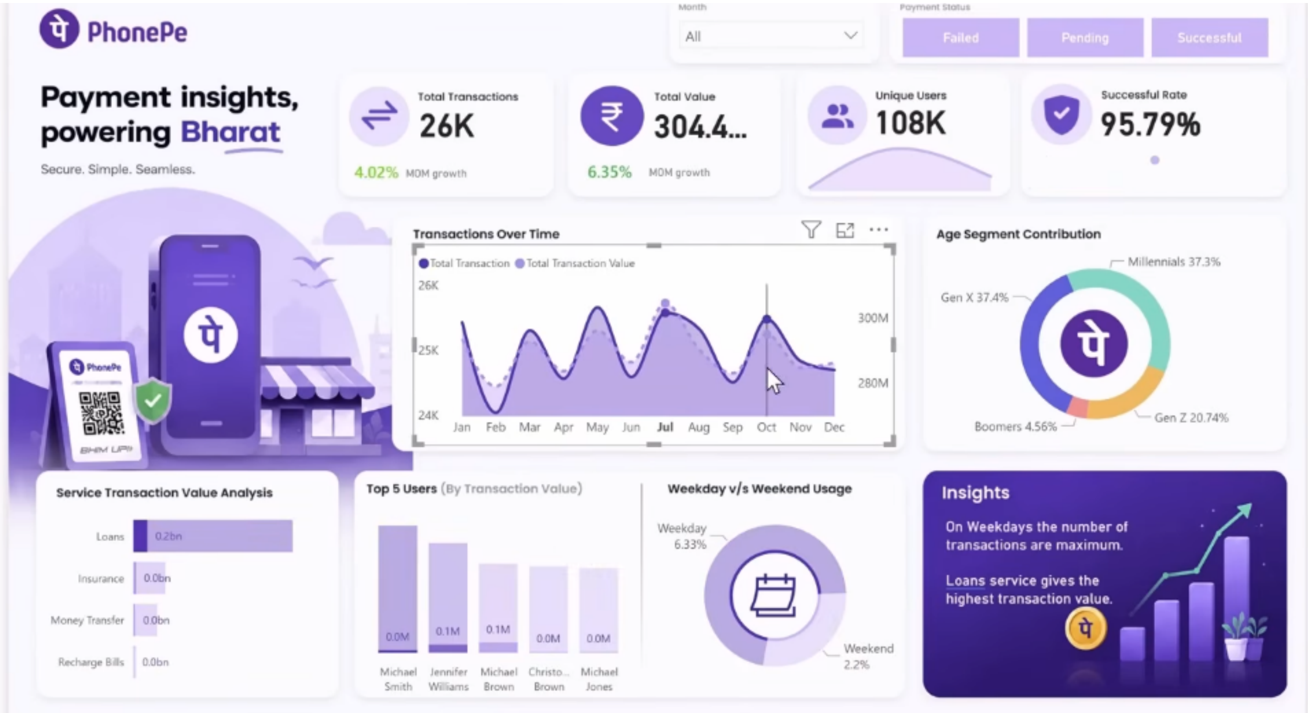

Fig 1. PhonePe Payment Insights dashboard — Page 1 (Power BI Desktop)

Skills Demonstrated

- Data modeling: star-schema style relationship between All_Transactions and All_Users on User_ID
- DAX measure design: CALCULATE, DIVIDE, DATEADD, WEEKDAY, COUNT, DISTINCTCOUNT and time-intelligence (MOM%) patterns
- Filter-context aware KPI cards that recalculate live from slicer selections (Month, Payment Status)
- Dynamic natural-language insights: text boxes bound to live measure values via Power BI's data-value insertion, not hardcoded strings
- Custom visual formatting: donut/line/bar chart theming, in-bar data labels, brand-matched color palette
- UX/report design: card layout, iconography, rounded insight panels, PhonePe brand styling

Feature Spotlight: Dynamic, Slicer-Responsive Insights

The dashboard's KPI cards and insight panels are not static snapshots — they are built on DAX measures, so every number and every generated sentence recalculates the instant a slicer is touched.

How it works

- Every KPI (Total Transactions, Total Value, Success Rate, etc.) is a DAX measure, not a hardcoded value — so it inherits whatever filter context is active on the page.
- Selecting **Failed**, **Pending**, or **Successful** on the Payment Status slicer filters All_Transactions[Payment_Status] before every measure evaluates, so cards like Total Transactions, Total Value and Success Rate instantly recompute for that status only.
- The purple 'Insights' panel uses Power BI's **Insert data value** feature: a live measure result (e.g. the top-performing Service by Total Transaction Value) is embedded directly inside a sentence, so the narrative text itself changes as filters change — it doesn't just relabel a chart.
- The same mechanism drives the Month slicer: switching months re-narrates the insights instead of just moving a bar.

Example DAX behind a filter-aware measure

```
Successful Transaction = CALCULATE([Total Transaction], All_Transactions[Payment_Status] =
"Successful")
Success Rate = DIVIDE([Successful Transaction], [Total Transaction])
```

Because Success Rate is a ratio of two CALCULATE-based measures rather than a fixed number, ticking the Payment Status slicer to “Failed” immediately shows the failure-rate equivalent for that slice of data — the card title stays the same, but the value and the accompanying insight text both re-evaluate live.

Data Model Summary

Table	Role
All_Transactions	Fact table — Transaction_ID, Amount, Service, Payment_Status, Date
All_Users	Dimension — User_ID, Age, Age Segment, Join_Date
Date_Table	Calendar table used for time-intelligence (MOM%) measures
Measures (2)	Central measures table holding all DAX calculations

Key measures

- Total Transaction Value = SUM(All_Transactions[Amount])
- Total Transaction = COUNT(All_Transactions[Transaction_ID])
- Successful Transaction = CALCULATE([Total Transaction], Payment_Status = "Successful")
- Success Rate = DIVIDE([Successful Transaction], [Total Transaction])
- Total Users = DISTINCTCOUNT(All_Users[User_ID])
- Trans Value MOM% = DIVIDE([Total Transaction Value] – [Trans Value PM], [Trans Value PM])

Prepared as a portfolio case study for the PhonePe Power BI Dashboard project — built end-to-end in Power BI Desktop, covering data modeling, DAX, and interactive report design.